

Digital Health M – Group Hands-on Project P3

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[watch our walkthrough video here](#)

Augmentation of Tabular Data for Boosting Accuracy

Second-cycle degree (LM) in Computer Engineering (cod. 5826)

Academic Year 2025/2026

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Project Objective

Classification Task
INPATIENT vs. OUTPATIENT

then

baseline Random Forest

SMOTE augmentation

Noise Injection augmentation

then

*evaluate **performance***



Dataset description

3000 patient records

FEATURES

SEX as CATEGORICAL VALUE

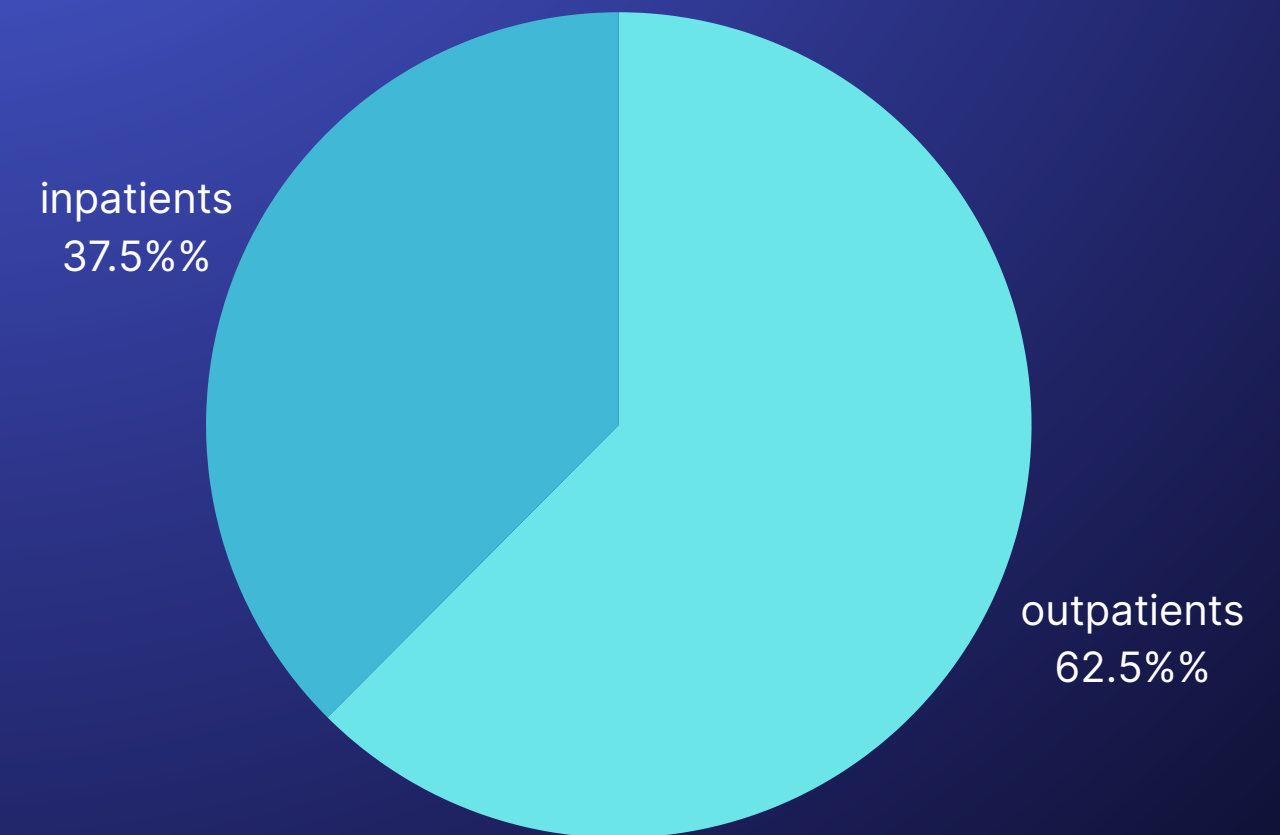
AGE and CBC as NUMERICAL VALUES

NO MISSING VALUES

TARGET

INPATIENTS

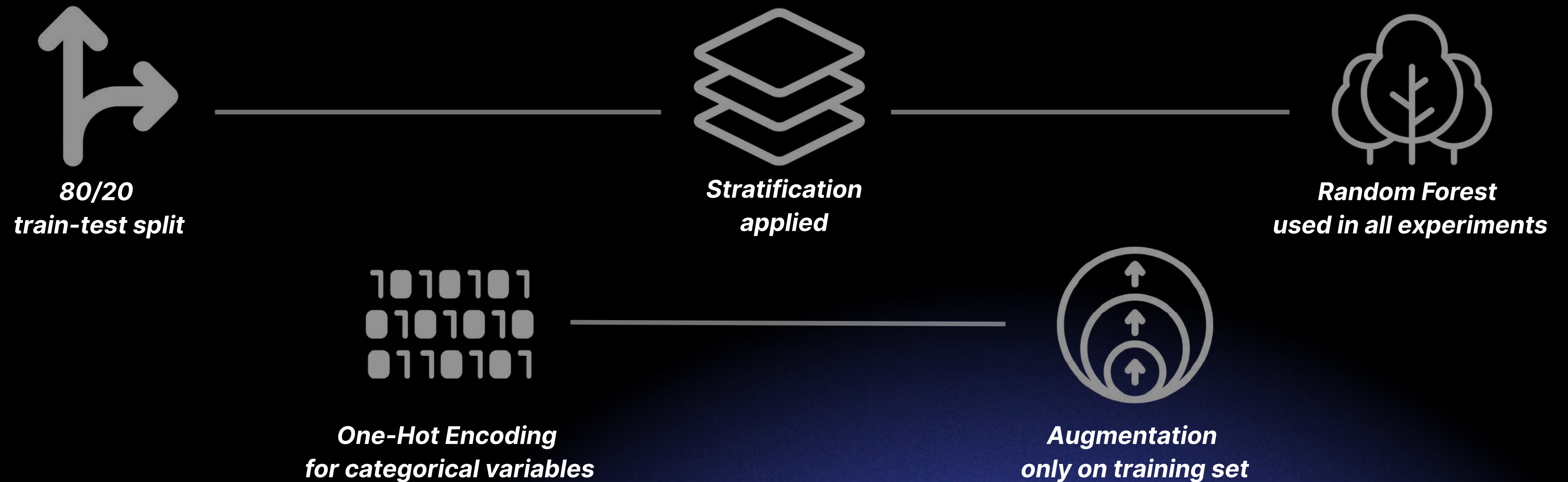
OUTPATIENTS



MODERATE IMBALANCE



experimental setup



baseline

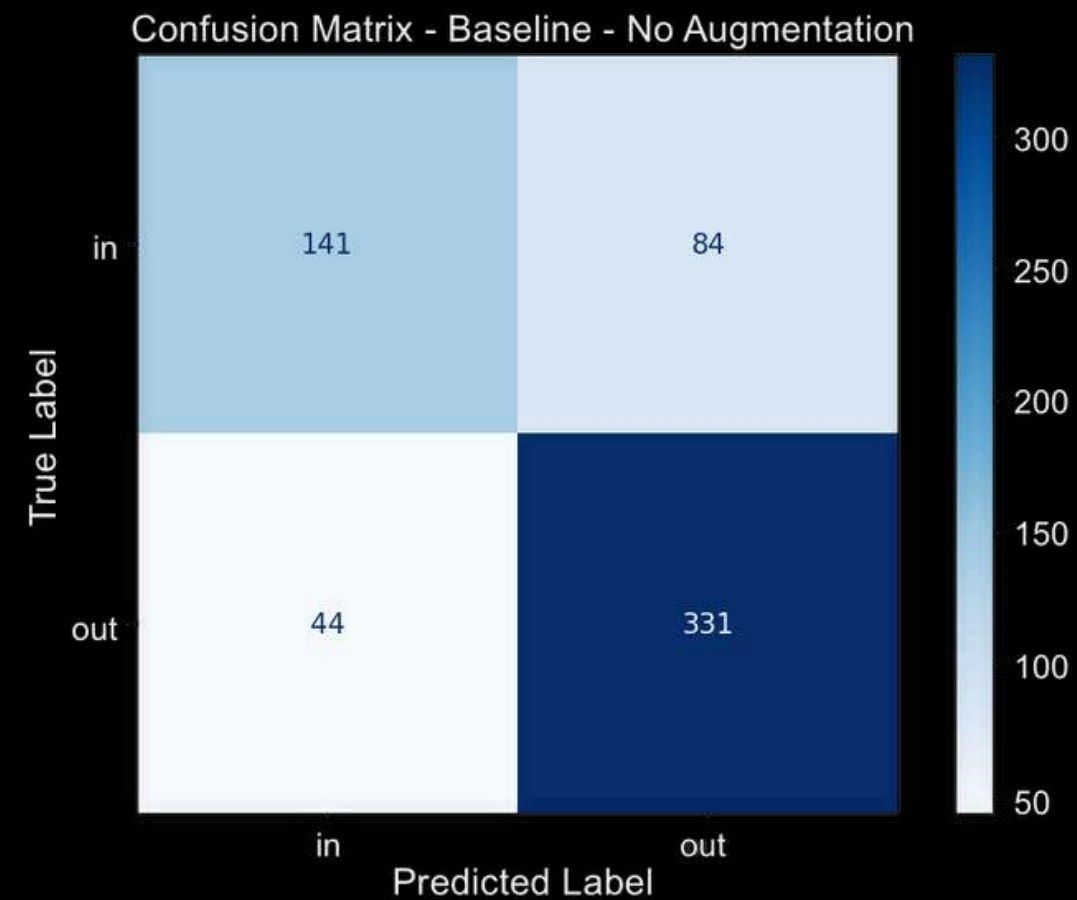


Table 5: Baseline model performance

Metric	Score
Accuracy	0.7867
Precision weighted	0.7843
Recall weighted	0.7867
F1 weighted	0.7817
Precision macro	0.7799
Recall macro	0.7547
F1 macro	0.7629

ACCURACY → 0.7867
BEST WEIGHTED F1-SCORE
STRONGEST OVERALL PERFORMANCE

SMOTE

SYNTHETIC MINORITY OVERSAMPLING TECHNIQUE
GENERATES SYNTHETIC MINORITY SAMPLES

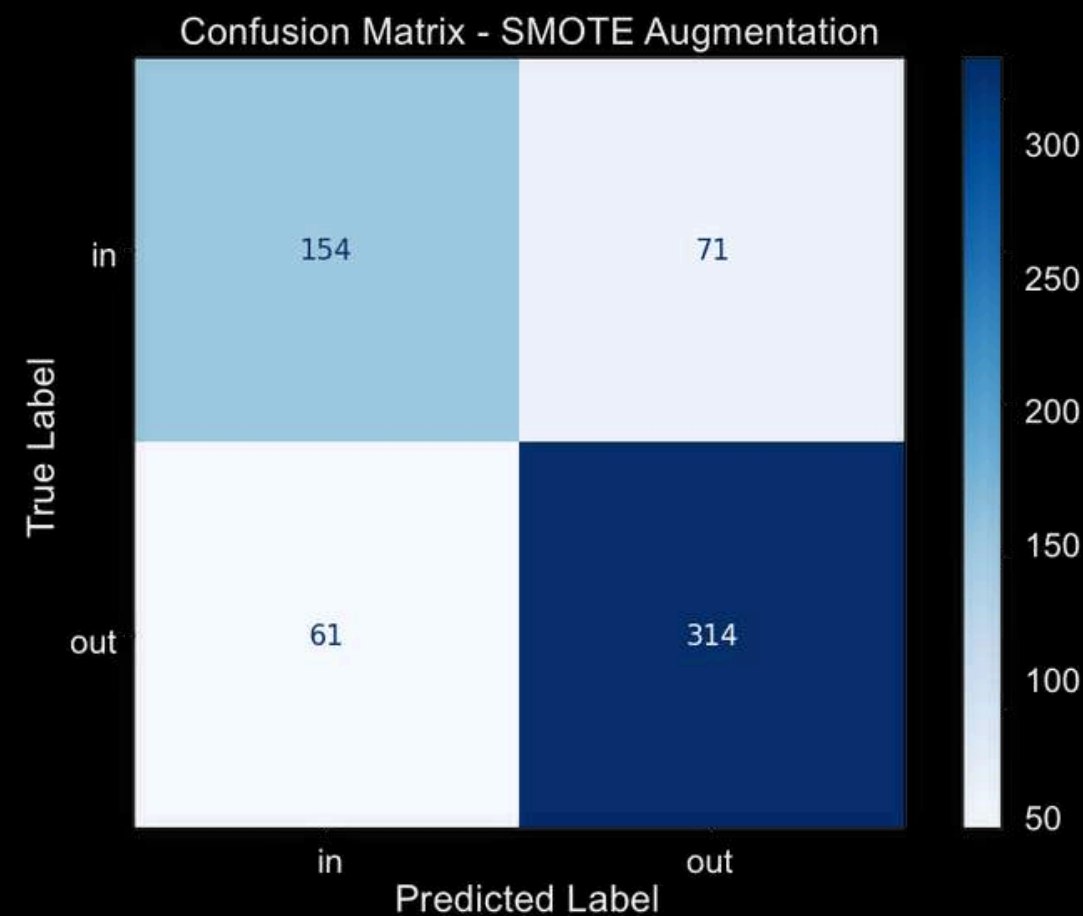


Table 6: SMOTE model performance

Metric	Score
Accuracy	0.7800
Precision weighted	0.7783
Recall weighted	0.7800
F1 weighted	0.7789
Precision macro	0.7659
Recall macro	0.7609
F1 macro	0.7632

GOAL
REDUCE BIAS TOWARD MAJORITY CLASS

SLIGHT IMPROVEMENTS IN MACRO METRICS

noise injection

SMALL RANDOM PERTURBATIONS APPLIED ONLY TO NUMERICAL FEATURES
SEX VARIABLE NOT MODIFIED

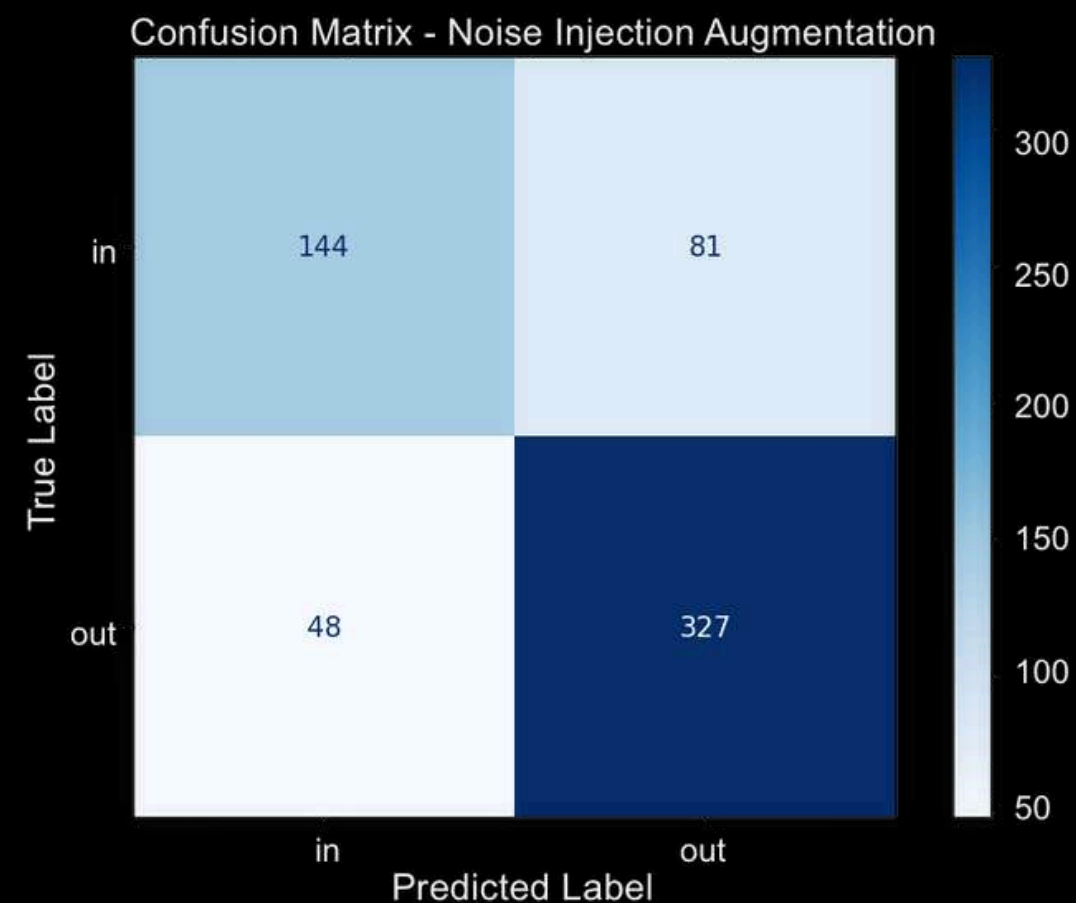


Table 7: Noise Injection model performance

Metric	Score
Accuracy	0.7850
Precision weighted	0.7822
Recall weighted	0.7850
F1 weighted	0.7810
Precision macro	0.7757
Recall macro	0.7560
F1 macro	0.7629

RESULTS CLOSE TO BASELINE

performance comparison

01 ——— 02 ——— 03

BASELINE

best accuracy
best weighted F1

SMOTE

best macro recall
best macro F1

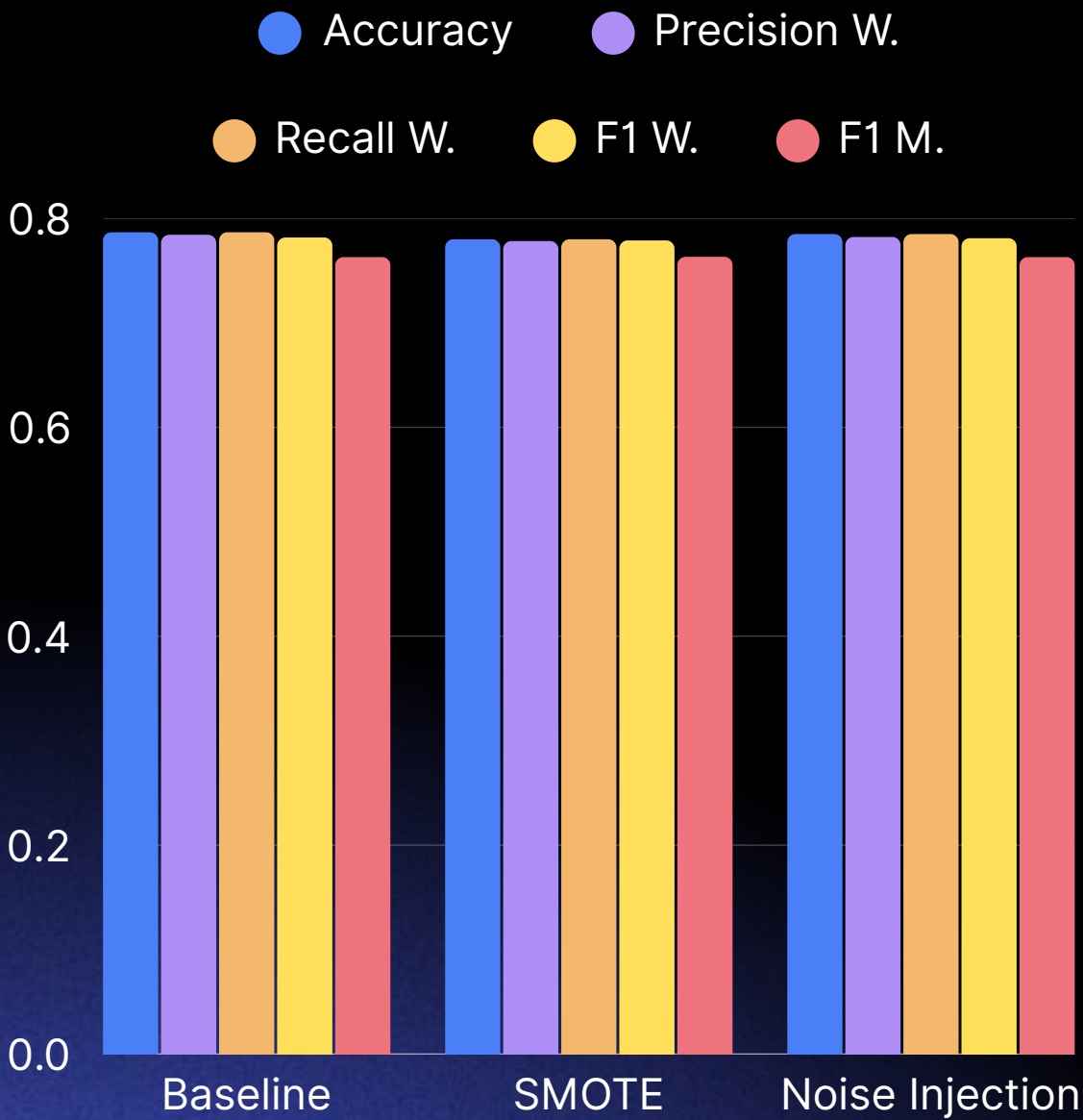
noise injection

similar to
baseline

Table 8: Performance comparison between baseline and augmented models

Model	Accuracy	Precision W.	Recall W.	F1 W.	Precision M.	Recall M.	F1 M.
Baseline – No Augmentation	0.7867	0.7843	0.7867	0.7817	0.7799	0.7547	0.7629
SMOTE Augmentation	0.7800	0.7783	0.7800	0.7789	0.7659	0.7609	0.7632
Noise Injection Augmentation	0.7850	0.7822	0.7850	0.7810	0.7757	0.7560	0.7629

baseline model achieved best accuracy and w. F1 score
SMOTE achieved best macro recall and macro F1
noise injection produced results close to the baseline



challenges



Clinical plausibility
of synthetic data



Correlated medical variables



Risk of **unrealistic** patient profiles



SMOTE
after One-Hot Encoding **limitation**

lessons learned



Augmentation
is **not always beneficial**



Metrics must be
interpreted **carefully**



Clinical meaning matters



Importance of **trustworthy**
synthetic data



Thank
You.